

Ali Reza Ibrahimzada

✉ alibrahimzada@gmail.com

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🌐 <https://alirezai.cs.illinois.edu/>

Research Interest

My research covers software engineering, machine learning, artificial intelligence, LLM agents, LLMs for code, and formal reasoning with the goal of modernizing and enhancing the reliability of large-scale software systems. To this end, I leverage program analysis and LLMs to automatically translate and validate repository-level code across different programming languages. So far, my work has focused on evaluating LLMs in code translation and investigating translation bugs [C4], source code decomposition for code translation [C5], and building end-to-end neuro-symbolic [C3] and agentic [C1] repository-level code translation and validation [P1, C2] tools.

Beside code translation, I have also enjoyed working on other problems in software engineering [C6, C7] and machine learning [J1, J2, J3].

Education

- 08/22 – 08/26  **Ph.D. Computer Science, University of Illinois Urbana-Champaign, IL, USA**
GPA: 4.00 / 4.00
Advisor: Reyhaneh Jabbarvand
Thesis: *Neuro-Symbolic Code Translation and Validation*
- 08/22 – 12/24  **M.Sc. Computer Science, University of Illinois Urbana-Champaign, IL, USA**
GPA: 4.00 / 4.00
Advisor: Reyhaneh Jabbarvand
Thesis: *Bridging the Gap between Testing and Debugging through Explainable Deep Oracles*
- 09/18 – 07/22  **B.Sc. Computer Engineering, Marmara University, Istanbul, Turkey**
GPA: 3.98 / 4.00
Advisor: Mehmet Kadir Baran
Thesis: *Depth Estimation of Stereo Images using Deep Learning*

Professional Experience

- 08/22 – Present  **Research Assistant, University of Illinois Urbana-Champaign, IL, USA**
Advised by: Reyhaneh Jabbarvand
Topic: Code LLMs for Source Code Generation, Understanding and Translation
Responsibilities:
- Performing research in the intersection of large language models and software engineering.
 - Evaluating large language models in code translation.
 - Building neuro-symbolic tools for unified translation and validation of repository-level code.
 - Creating agentic architectures for translation validation and repair.
 - Developing agentic workflows for end-to-end whole-repository translation and validation.

Professional Experience (continued)

- 05/25 – 08/25  **Applied Scientist Intern**, Amazon Web Services, VA, USA
Hosted by: **Brandon Paulsen** and **Joey Dodds**
Topic: LLM Agents for Autonomous Translation Validation and Repair
Responsibilities:
- Building MatchFixAgent, a language-agnostic and agentic technique for translation validation and repair.
 - Creating benchmarks from existing techniques and evaluating MatchFixAgent.
 - Performing cost analysis of MatchFixAgent against existing language-specific techniques.
- 05/24 – 08/24  **Research Intern**, IBM Research, NY, USA
Hosted by: **Saurabh Sinha** and Raju Pavuluri
Topic: End-to-End Repository-level Code Translation and Validation
Responsibilities:
- Developing AlphaTrans, the first end-to-end neuro-symbolic tool for repository-level code translation and validation.
 - Creating a benchmark of real-world Java repositories and evaluating AlphaTrans on them.
 - Developing a language interoperability approach for validating translations in isolation.
- 05/23 – 08/23  **Research Intern**, IBM Research, NY, USA
Hosted by: **Saurabh Sinha** and Raju Pavuluri
Topic: Evaluating LLMs in Code Translation and Bug Analysis
Responsibilities:
- Performing a large-scale empirical study on LLM-based code translation.
 - Performing a large-scale human study on translation bugs generated by LLMs.
 - Comparing LLMs with transpilers in code translation.
- 05/21 – 08/22  **Research Intern**, University of Illinois Urbana-Champaign, IL, USA
Hosted by: **Reyhaneh Jabbarvand**
Topic: Neural and Interpretable Test Oracles using Deep Learning
Responsibilities:
- Building SEER, a deep neural model for test oracle problem.
 - Evaluating SEER on the Defects4J benchmark.
 - Comparing SEER with traditional mutation-based techniques.
- 06/19 – 02/22  **Research Intern**, Istanbul Technical University, Istanbul, Turkey
Hosted by: **Ali Cakmak**
Topic: Machine Learning
Responsibilities:
- Research on using machine learning techniques for scalable classification of living organisms.
 - Developing machine learning techniques for predicting future COVID mutants.
 - Developing tools for predicting student success using machine learning.

Research Publications

Preprints

- [P1] K. Ke, **A. R. Ibrahimzada**, R. Pan, S. Sinha, and R. Jabbarvand, “Advancing automated in-isolation validation in repository-level code translation,” 2025.  URL: <https://arxiv.org/abs/2511.21878>.

Conference Proceedings

- [C1] **A. R. Ibrahimzada**, B. Paulsen, D. Kroening, and R. Jabbarvand, "Recodeagent: A multi-agent workflow for language-agnostic translation and validation of large-scale repositories," in *Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering*, ser. ASE '26, Munich, Germany, 2026.  URL: <http://arxiv.org/abs/2604.07341>.
- [C2] **A. R. Ibrahimzada**, B. Paulsen, R. Jabbarvand, J. Dodds, and D. Kroening, "Matchfixagent: Language-agnostic autonomous repository-level code translation validation and repair," in *International Conference on Machine Learning*, PMLR, 2026.  URL: <https://openreview.net/forum?id=MuyXpH3GL1>.
- [C3] **A. R. Ibrahimzada**, K. Ke, M. Pawagi, M. S. Abid, R. Pan, S. Sinha, and R. Jabbarvand, "Alphatrans: A neuro-symbolic compositional approach for repository-level code translation and validation," *FSE*, vol. 2, New York, NY, USA: Association for Computing Machinery, Jun. 2025.  DOI: [10.1145/3729379](https://doi.org/10.1145/3729379).
- [C4] R. Pan*, **A. R. Ibrahimzada***, R. Krishna, D. Sankar, L. P. Wassi, M. Merler, B. Sobolev, R. Pavuluri, S. Sinha, and R. Jabbarvand, "Lost in translation: A study of bugs introduced by large language models while translating code," in *Proceedings of the IEEE/ACM 46th International Conference on Software Engineering*, ser. ICSE '24, Lisbon, Portugal: Association for Computing Machinery, 2024, ISBN: 9798400702174.  DOI: [10.1145/3597503.3639226](https://doi.org/10.1145/3597503.3639226).
- [C5] **A. R. Ibrahimzada**, "Program decomposition and translation with static analysis," in *Proceedings of the 2024 IEEE/ACM 46th International Conference on Software Engineering: Companion Proceedings*, ser. ICSE-Companion '24, Lisbon, Portugal: Association for Computing Machinery, 2024, pp. 453–455, ISBN: 9798400705021.  DOI: [10.1145/3639478.3641226](https://doi.org/10.1145/3639478.3641226).
- [C6] **A. R. Ibrahimzada**, Y. Varli, D. Tekinoglu, and R. Jabbarvand, "Perfect is the enemy of test oracle," in *Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, ser. ESEC/FSE 2022, Singapore, Singapore: Association for Computing Machinery, 2022, pp. 70–81, ISBN: 9781450394130.  DOI: [10.1145/3540250.3549086](https://doi.org/10.1145/3540250.3549086).
- [C7] **A. R. Ibrahimzada**, Y. Chen, R. Rong, and R. Jabbarvand, "Challenging bug prediction and repair models with synthetic bugs," in *2025 IEEE International Conference on Source Code Analysis & Manipulation (SCAM)*, 2025, pp. 133–144.  DOI: [10.1109/SCAM67354.2025.00021](https://doi.org/10.1109/SCAM67354.2025.00021).

Journal Articles

- [J1] **A. R. Ibrahimzada**, K. Kosif, A. S. Gulsen, Y. S. Yaglica, and A. Cakmak, "Predicting student grades via adaptive multi-level learning models," *Scientific Reports*, 2026.  URL: <https://doi.org/10.1038/s41598-026-55750-z>.
- [J2] A. Cakmak, H. Ayaz, S. Arıkan, **A. R. Ibrahimzada**, Ş. Demirkol, D. Sönmez, M. T. Hakan, S. T. Sürmen, C. Horozoglu, M. B. Doğan, Ö. Küçükhusayin, C. Cacına, B. Kıran, Ü. Zeybek, M. Baysan, and İ. Yaylım, "Predicting the predisposition to colorectal cancer based on snp profiles of immune phenotypes using supervised learning models," *Medical & Biological Engineering & Computing*, vol. 61, no. 1, pp. 243–258, Jan. 2023, ISSN: 1741-0444.  DOI: [10.1007/s11517-022-02707-9](https://doi.org/10.1007/s11517-022-02707-9).
- [J3] G. N. Sohsah, **A. R. Ibrahimzada**, H. Ayaz, and A. Cakmak, "Scalable classification of organisms into a taxonomy using hierarchical supervised learners," *Journal of Bioinformatics and Computational Biology*, vol. 18, no. 05, p. 2 050 026, 2020, PMID: 33125294.  DOI: [10.1142/S0219720020500262](https://doi.org/10.1142/S0219720020500262). eprint: <https://doi.org/10.1142/S0219720020500262>.

Thesis

- [T1] **A. R. Ibrahimzada**, *Neuro-symbolic code translation and validation*, 2026.
- [T2] **A. R. Ibrahimzada**, *Bridging the gap between testing and debugging through explainable deep oracles*, 2024.

Awards

- 2024
 - 📌 **Nomination for the Two Sigma PhD Fellowship**, University of Illinois Urbana-Champaign
 - 📌 **3rd in Student Research Competition @ ICSE'24**, ACM
 - 📌 **Travel Grant to attend ICSE'24 in Lisbon, Portugal**, NSF
- 2022
 - 📌 **Travel Grant to attend ESEC/FSE'22 in Singapore, Singapore**, ACM
 - 📌 **Valedictorian of Class of 2022**, Marmara University
 - 📌 **Ray Ozzie Computer Science Fellowship**, University of Illinois Urbana-Champaign
 - 📌 **Best Senior Graduation Project of the Year**, Marmara University
- 2019
 - 📌 **Academic Achievement Scholarship**, Marmara University
- 2017
 - 📌 **Valedictorian of High School**, KEN
- 2015
 - 📌 **Travel Grant to attend QUEST'15 in Lucknow, India**, KEN

Talks

- ASE'26
 - 📌 **ReCodeAgent: A Multi-agent Workflow for Language-Agnostic Translation and Validation of Large-Scale Repositories**, Munich, Germany
- ICML'26
 - 📌 **MatchFixAgent: Language-Agnostic Autonomous Repository-Level Code Translation Validation and Repair**, Seoul, South Korea
- SCAM'25
 - 📌 **Challenging Bug Prediction and Repair Models with Synthetic Bugs**, Auckland, New Zealand
- FSE'25
 - 📌 **AlphaTrans: A Neuro-Symbolic Compositional Approach for Repository-Level Code Translation and Validation**, Trondheim, Norway
- ICSE'24
 - 📌 **Lost in Translation: A Study of Bugs Introduced by Large Language Models while Translating Code**, Lisbon, Portugal
- FSE'22
 - 📌 **Perfect Is the Enemy of Test Oracle**, Singapore, Singapore

Skill Stack

- Programming Languages
 - 📌 Python, Java, JavaScript, Rust, C, OCaml
- Common
 - 📌 Docker, Vim, Git, SQL
- Machine Learning
 - 📌 PyTorch, TensorFlow, vLLM, Codex, Claude Code

Service

- Program Committee
 - 📌 ICLR'27 (San Francisco, USA), NeurIPS'26 (Sydney, Australia), ICML'26 (Seoul, South Korea), ICLR'26 (Rio de Janeiro, Brazil), ICLR'25 (Singapore, Singapore), MSR'25 (Ottawa, Canada), MSR'24 (Lisbon, Portugal)
- Artifact Evaluation
 - 📌 ICSE'25 (Ottawa, Canada)
- Organizing
 - 📌 ReCode@ICSE'27 (Dublin, Ireland), ReCode@ICSE'26 (Rio de Janeiro, Brazil)

Teaching Experience

- Fall 2025
 - 📌 CS 421 - Programming Languages and Compilers w/ Elsa Gunter
- Spring 2025
 - 📌 CS 447 - Natural Language Processing w/ Julia Hockenmaier

References

Reyhaneh Jabbarvand

Assistant Professor

University of Illinois Urbana-Champaign

201 N Goodwin Ave, 61801, Urbana, IL

✉ reyhaneh@illinois.edu

Darko Marinov

Professor

University of Illinois Urbana-Champaign

201 N Goodwin Ave, 61801, Urbana, IL

✉ marinov@illinois.edu

Saurabh Sinha

Principal Research Scientist

IBM Research

1101 Kitchawan Rd, 10598, Yorktown Heights, NY

✉ sinhas@us.ibm.com

Brandon Paulsen

Senior Applied Scientist

Amazon Web Services

1400 S Eads St, 22202, Arlington, VA

✉ bpaulse@amazon.com